



Database Foundations

2-2

Conceptual and Physical Data Models



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Objectives

- This lesson covers the following objectives:
 - Describe a conceptual data model
 - Describe a logical data model
 - Describe a physical data model
 - Analyze the similarities and differences between conceptual and physical data models



What Is a Conceptual Model? 1 of 2

- Captures the functional and informational needs of a business
- Is based on current needs, but may reflect future needs
- Addresses the needs of a business (what is conceptually ideal), but does not address its implementation (what is physically possible)



What Is a Conceptual Model? 2 of 2

- Identifies :
 - important entities (objects that become tables in database)
 - relationships among entities
- Does not specify :
 - attributes (objects that become columns or fields in database)
 - unique identifiers (attribute that becomes primary key in database)



What Is a Logical Model?

- Includes all entities and relationships among them
- Is called an entity relationship model (ERM)
- Is illustrated in an ERD
- Specifies all attributes and UUIDs for each entity
- Determines attribute optionality
- Determines relationship optionality and cardinality *



A logical data model describes the data in as much detail as possible, without regard to how it will be physically implemented in the database. It is normally derived from a conceptual data model.

* We will discuss optionality and cardinality later in the course

What Is a Physical Model?

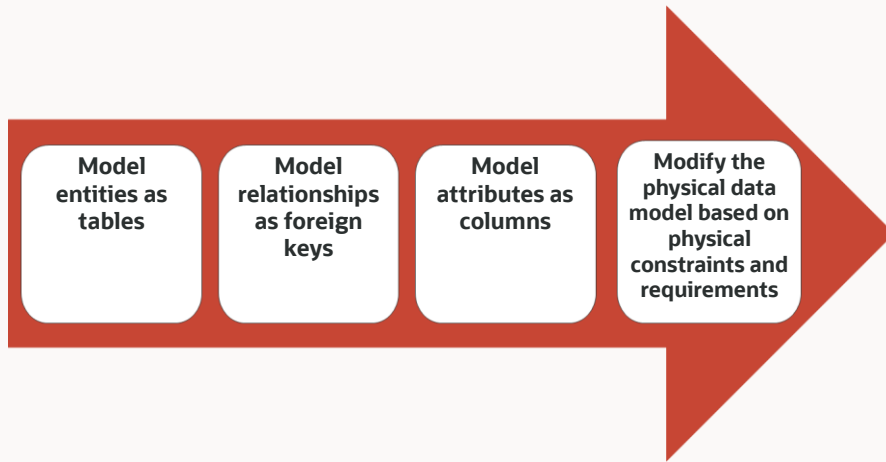
- Is an extension to a logical data model
 - Defines table definitions, data types, and precision
 - Identifies views, indexes, and other database objects *
- Describes how the objects should be implemented in specific database
- Shows all table structures, including columns, primary keys, and foreign keys



Physical modeling deals with the conversion of the logical data model to a relational database model. Each relational model can have one or more physical models, one for each RDBMS that you deploy to.

* We will discuss views, indexes and other database objects later in the course.

Steps to Create a Physical Data Model



Conceptual and Physical Models

- The art of planning, developing, and communicating produces a desired outcome
- Data modeling is the process of capturing the important concepts and rules that shape a business and depicting them visually in a diagram
- This diagram becomes the blueprint for designing the physical thing
- The client's dream (conceptual model) becomes a physical reality (physical model)



- The conceptual model is concerned with the real-world view and understanding of data. The physical model specifies how it will be executed in a particular database management instance.

Conceptual/Logical Model: Case Scenario



Faculty

Matt, I would like you to create a simplified library database to store the book details in our department.

Sure. I will start by identifying the entities and attributes and their relationships.



Matt



Case Scenario: Creating a Conceptual Model

- A Conceptual data model documents important entities and how they relate to each other

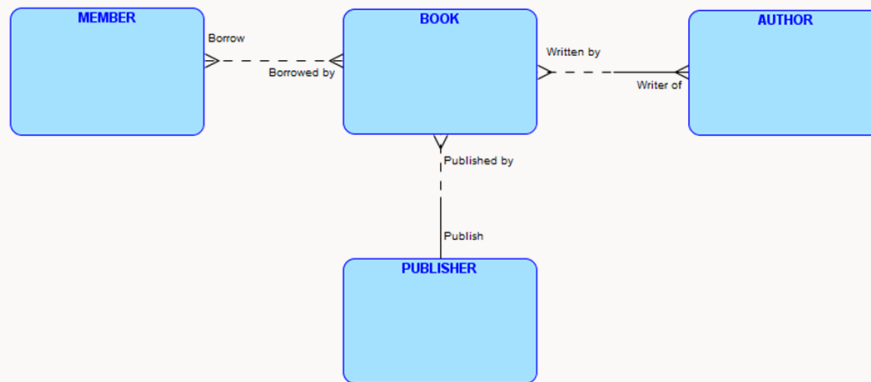


Diagram was created using Oracle SQL Data Modeling tool which will be covered in Section 4 of course material.

Case Scenario: Creating a Logical Model

- A logical data model documents the information requirements of the business

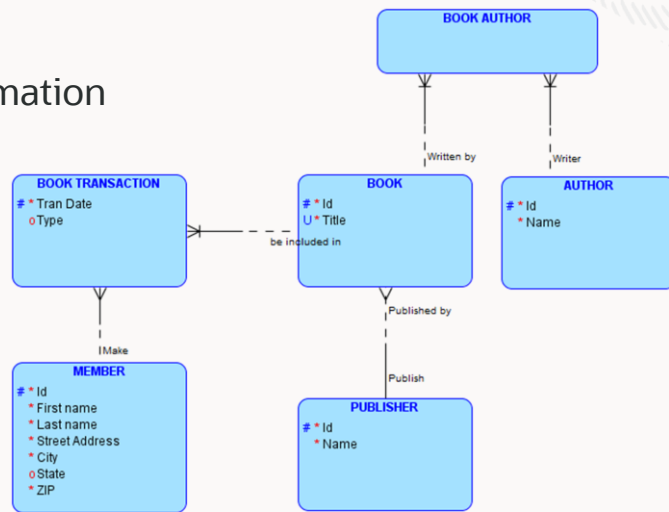


Diagram was created using Oracle SQL Data Modeling tool which will be covered in Section 4 of course material.

Physical Model: Case Scenario



Faculty

Matt, I want to know the specifications for all tables and columns required in the simplified library database.

Sure. I will convert the entities and their attributes into tables and columns.

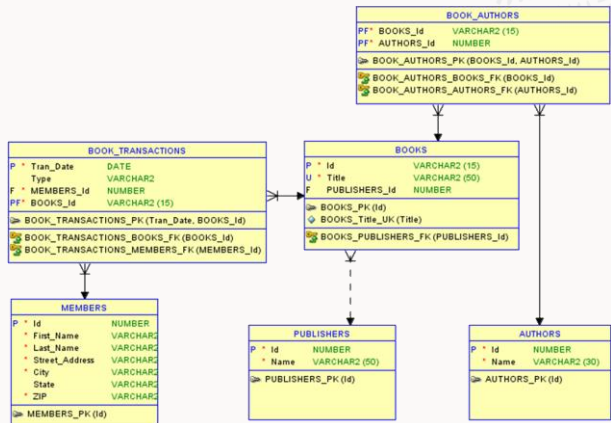


Matt



Case Scenario: Creating a Physical Model

- Modeling Performed
 - Entities → Tables
 - Attributes → Columns
 - Relationships → Foreign Keys



- Diagram was created using Oracle SQL Data Modeling tool which will be covered in Section 4 of course material.

Summary

- In this lesson, you should have learned how to:
 - Describe a conceptual model
 - Describe a logical model
 - Describe a physical model
 - Analyze the similarities and differences between conceptual and physical data models



